

the world, *Digit* hazards a guess that we'll be chatting up to and making jokes with our computers quite routinely within the next decade. And then, perhaps, they may just end up replacing dogs as man's best friend.

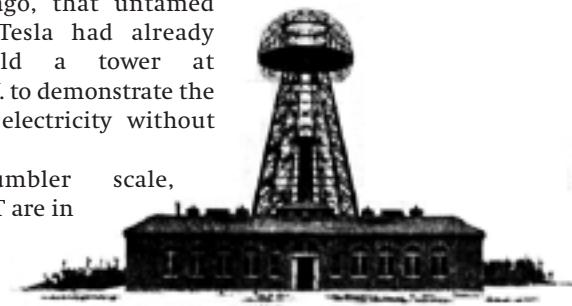
7. WIRELESS ELECTRICITY

Power's In The Air

Tired Of Being Wired

If phones, mice and keyboards could get wireless, why not everything else? In fact, about a hundred years ago, that untamed genius, Nikola Tesla had already begun to build a tower at Wardencllyffe, N.Y. to demonstrate the transmission of electricity without the use of wires.

On a humbler scale, researchers at MIT are in the process of repeating the experiment with their own ideas and less ostentatious techniques.



The Wardencllyffe Tower was never finished

WiTricity

Marin Soljacic, Assistant Professor of Physics at the MIT, has spent a considerable number of years trying to figure out how to transmit power without cables. Radio waves lose too much energy during their passage through the air and lasers are constrained by the necessity of line-of-sight. Soljacic decided to use Resonant Coupling, in which two objects vibrating at the same frequency can exchange energy without harming things around them. He used magnetic resonance and along with his colleagues Jon Hoannopoulos and Peter Fisher, succeeded in lighting up a 60 watt bulb two metres away. What they did was this: two resonant copper coils were tied to dangle from the ceiling, two meters away from each other. Both were tuned to the same frequency and one had a light bulb attached to it. When current was made to pass through one coil, it created a magnetic field and the other resonated, generating an electric current. And then there was light. The experiment succeeded in spite of a thin screen being placed between the two copper coils.

And If This Comes Through...

One of the most obvious results is that we won't have dozens of cables to trip over in our offices and rooms. Primarily, the aim of this research team is to achieve a cable-free environment wherein your laptops PDAs and mobile phones could charge themselves (with all the electricity floating around) and even,

maybe, get rid of the batteries that are so much an essential part of our portable devices today. Magnetic fields interact very weakly with biological organisms and this little fact makes it infinitely safer for us. While this experiment happened about a year ago, the team is still hard at work trying to use other materials so as to increase the efficiency of the transfer of power from 50 per cent to 80 per cent. Once that happens, both, the industry as well as individuals will grab hold of it and never let go.

8. TABLE-SCREENS

Scribbling On The Desk

No More Dead Wood

There isn't a student alive who hasn't sometime scribbled his name (or a caricature of his prof.) on his school desk. How much more exciting it would have been if your desk was actually a Graphical User Interface! Experts at Durham University are aiming for just that with their 'Smart-Desk' initiative.

Interactive Surfaces

The Active Learning in Computing department at Durham University, UK is designing interactive multi-touch desks at their TEL (Technology-Enhanced Learning Research Group), hoping to replace the traditional desk with cell-phone-like touch-screens which can act like a multi-touch whiteboard, a keyboard, and mobile screen that several students can use at the same time. Dr Liz Burd and her team have linked up with private enterprises to design software that will enable all these surfaces to be networked and connected to a main smartboard. The computer becomes a part of the desk.

And If This Comes Through...

Instant visual displays of topics being discussed, on-screen interactive mathematics, group efforts in problem-solving and more involvement of students in the task at hand — the possible advantages of the smart desk to teachers and students seem infinite. Students who tend to isolate themselves or resist participation in class would be coerced to interact. Teamwork would be a natural consequence with multiple users on single screens. Each student could be presented a task or problem according to his or her individual capacities. More active and creative tasks would replace passive listening. The team aims to fill all schools in the UK with these desks within a decade and keeping in mind the pace of technological advance in India, we should see at least some of the schools in the country doing the same in the near future.

9. WRAP-AROUND COMPUTERS

Open New Folder

Why Should Screens Be Flat?

Flat screens are in. But what if we could have screens folded or curved around any surface that was convenient to reach? What if animated billboards could be folded round the pole of a street light? What if you could watch your favourite movie by stretching the screen over the back of a chair?

Bendable and flexible electronics are already all over the news. Trouble is, most of them cannot be tied up or wrapped around uneven surfaces or complicated shapes. Nanotechnology (what else?) has the answer.

Elastic Computing

Takao Someya, professor of engineering at the University of Tokyo, along with his team of researchers has added carbon nanotubes to a polymer with high elasticity to make a conductive material. They then used this to connect organic transistors in a stretchable electronic circuit. To induce conductivity in this material, Someya and his team combined several single-walled carbon nanotubes with an ionic liquid, which took the form of a black paste. This substance was then added to liquid polymer, which was dried after being poured into a cast. This material could be used to make an 'electronic skin' for robots. As a result, the nanotubes were evenly spread in the material and these worked to form a network that permitted electrical signals to pass through in a controlled manner. To make this material more stretchable, it was perforated into a net and then coated with a material with a silicone base.

In a paper published in 'Science' magazine, Someya reported that the material has the highest conductivity among soft materials in the world. Besides, the material is able to stretch up to about 134 percent of its original shape.

And If This Comes Through...

Mass production of nanotubes would, in turn, assist in the bulk production of these elastic conductors. The new material could be used to make displays, actuators or computers. With foldable keyboards already in the market for quite a while, a stretchable screen would make carrying around your laptop infinitely easier. It won't be long before you'll reach into your pocket for a handkerchief and pull out your PC. Look before you sneeze.

10. ULTRA-COMPRESSED MUSIC FILES

Micro-MP3

Just Forty Thousand Songs?

The Apple iPod (160 GB) can hold about forty thousand songs and, yes, people are buying it. As the capacity of MP3 players increase, strangely, our list of must-have favourite songs also expands exponentially. It doesn't matter how many songs you've got with you, the song you want is always elsewhere. So for those of you out there, who, have a million favourite tunes, never fear, Rochester's here. Again.

Zipped A Thousand Times

Researchers at the University of Rochester have succeeded in digitally reproducing a piece of music in a file that is almost 1,000 times smaller than a regular MP3 file. Mark Bocko, professor of electrical and computer engineering, and his team announced this achievement, at the International Conference on Acoustic Speech and Signal Processing in Las Vegas, held on April 1, 2008.

Even though the results are not perfect, they are almost so. The team took as a sample, a short musical piece, a 20-second solo on a clarinet and compressed it to less than a kilobyte. The file was then replayed by a combination of physics and the knowledge of how a clarinet works. They fed into the computer everything about clarinet playing — including the movement of the fingers, the pressure on the mouthpiece, etc. — to create a virtual clarinet based on real-world dimensions and parameters. They then made a virtual clarinet player for this virtual clarinet by feeding in a model that tells the computer how the human player interacts with the instrument including the fingerings, the force of breath, and the pressure of the player's lips to determine how they would affect the response of the virtual clarinet.

And If This Comes Through...

Not only does this imply the possibility of ultra-compressed music files but also the incredible prospect of recording the performer along with performance. Once a computer figures out the typical style of a player, his every breath and movement, it could play a tune much after the player's gone. According to Professor Bocko, improvement in quality is inevitable as the algorithms become more accurate and acoustic measurements are further perfected. The day won't be far when your cell phone will hold all the music ever produced in the whole wide world — and the movies too. ■